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Haloalkanes and Haloarenes

1. Haloalkanes

- **Definition:** Alkanes with one or more H atoms replaced by halogen atoms (X = F, Cl, Br, I).
- **Nomenclature:** Common names (alkyl halides), IUPAC names (haloalkanes).
- **Preparation:**
  - From Alcohols: $ROH + HX \rightarrow RX + H_2O$ (Lucas reagent for 3° alcohols)
  - From Alkenes: $R-CH=CH_2 + HX \rightarrow R-CH(X)-CH_3$ (Markovnikov's rule)
  - Free radical halogenation of alkanes: $CH_4 + Cl_2 \xrightarrow{hv} CH_3Cl + HCl$
  - Halogen exchange: Finkelstein reaction (RI), Swarts reaction (RF).
- **Physical Properties:**
  - Boiling points: $RI > RBr > RCl > RF$. Increases with molecular mass.
  - Density: $RI > RBr > RCl$.
  - Solubility: Insoluble in water, soluble in organic solvents.
- **Chemical Reactions:**
  - **Nucleophilic Substitution (S_N1, S_N2):**
    - S_N2: Concerted, one step, $2^\circ > 1^\circ > 3^\circ$ (steric hindrance), inversion of configuration.
    - S_N1: Two steps, carbocation intermediate, $3^\circ > 2^\circ > 1^\circ$ (carbocation stability), racemization.
    - Reactivity: $RI > RBr > RCl > RF$.
  - **Elimination Reactions (E1, E2):** Dehydrohalogenation (Saytzeff's rule).
  - **Reaction with Metals:** Wurtz reaction ($R-X+2Na+X-R \rightarrow R-R+2NaX$), Grignard reagent ($RMgX$).

2. Haloarenes

- **Definition:** Halogen atom directly attached to an aromatic ring.
- **Preparation:**
  - Halogenation of benzene: $C_6H_6 + Cl_2 \xrightarrow{FeCl_3} C_6H_5Cl + HCl$
  - Sandmeyer reaction: $ArN_2^+Cl^- \xrightarrow{Cu_2X_2} ArX + N_2$
  - Gattermann reaction: $ArN_2^+Cl^- \xrightarrow{Cu/HX} ArX + N_2$
- **Physical Properties:** Higher melting and boiling points than haloalkanes.
- **Chemical Reactions:**
  - **Nucleophilic Substitution:** Difficult due to resonance stabilization, sp^2 carbon, and repulsion.
  - **Electrophilic Substitution:** Halogen is deactivating but o,p-directing.
    - Halogenation, Nitration, Sulphonation, Friedel-Crafts alkylation/acylation.
  - **Reaction with Metals:** Wurtz-Fittig reaction ($Ar-X+2Na+X-R \rightarrow Ar-R+2NaX$), Fittig reaction ($Ar-Ar$).

Alcohols, Phenols and Ethers

1. Alcohols

- **Definition:** Organic compounds containing -OH group attached to an alkyl group.
- **Classification:** Primary (1°), Secondary (2°), Tertiary (3°).
- **Preparation:**
  - From Alkenes: Hydration (acid-catalyzed), hydroboration-oxidation.
  - From Carbonyl compounds: Reduction of aldehydes/ketones ($LiAlH_4$, $NaBH_4$), Grignard reagent.
  - From Carboxylic acids and esters: Reduction ($LiAlH_4$).
- **Physical Properties:** Higher boiling points due to H-bonding. Soluble in water (short chain).
- **Chemical Reactions:**
  - **Acidity:** $H_2O > 1^\circ > 2^\circ > 3^\circ$ (due to +I effect).
  - **Reaction with HX:** $ROH + HX \rightarrow RX + H_2O$ (Lucas test).
  - **Dehydration:** $ROH \xrightarrow{conc.H_2SO_4} Alkene$ (Saytzeff's rule).
  - **Oxidation:** $1^\circ \rightarrow$ Aldehyde $\rightarrow$ Carboxylic acid; $2^\circ \rightarrow$ Ketone; 3° no oxidation.
  - **Esterification:** $ROH + R'COOH \rightleftharpoons R'COOR + H_2O$.

2. Phenols

- **Definition:** -OH group directly attached to an aromatic ring.
- **Preparation:**
  - From Haloarenes: Dow's process ($C_6H_5Cl \xrightarrow{NaOH,623K} C_6H_5ONa \xrightarrow{H^+} C_6H_5OH$).
  - From Benzene Sulphonic acid: $C_6H_5SO_3H \xrightarrow{NaOH} C_6H_5ONa \xrightarrow{H^+} C_6H_5OH$.
  - From Diazonium salts: $ArN_2^+Cl^- \xrightarrow{H_2O,\Delta} ArOH + N_2 + HCl$.
  - From Cumene: Cumene $\xrightarrow{O_2}$ Cumene hydroperoxide $\xrightarrow{H_3O^+}$ Phenol + Acetone.
- **Physical Properties:** Higher boiling points, less soluble in water than alcohols.
- **Chemical Reactions:**
  - **Acidity:** More acidic than alcohols due to resonance stabilization of phenoxide ion. e^- withdrawing groups increase acidity.
  - **Electrophilic Substitution:** -OH is activating and o,p-directing.
    - Nitration, Halogenation (Bromination), Sulphonation.
  - **Kolbe's Reaction:** Salicylic acid formation.
  - **Reimer-Tiemann Reaction:** Salicylaldehyde formation.
  - **Reaction with Zinc Dust:** $C_6H_5OH + Zn \rightarrow C_6H_6 + ZnO$.
  - **Oxidation:** With chromic acid gives benzoquinone.

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3. Ethers

- **Definition:** $R-O-R'$ where R and R' are alkyl or aryl groups.
- **Classification:** Symmetrical ($R=R'$) or Unsymmetrical ($R \neq R'$).
- **Preparation:**
  - **Williamson Synthesis:** $R-X + NaOR' \rightarrow R-O-R' + NaX$ (1° alkyl halide preferred).
  - Dehydration of alcohols: $2ROH \xrightarrow{H_2SO_4,413K} R-O-R + H_2O$.
- **Physical Properties:** Lower boiling points than parent alcohols (no H-bonding). Soluble in water (short chain).
- **Chemical Reactions:**
  - **Cleavage by HI/HBr:** $R-O-R' + HX \rightarrow RX + R'OH$ (SN2 mechanism, smaller alkyl group forms halide).
  - **Electrophilic Substitution (for aromatic ethers):** -OR is activating and o,p-directing.

Aldehydes, Ketones and Carboxylic Acids

1. Aldehydes and Ketones

- **Definition:** Contain the carbonyl group ($C=O$). Aldehydes ($RCHO$), Ketones ($RCOR'$).
- **Preparation:**
  - Oxidation of alcohols ($1^\circ \rightarrow$ Aldehydes, $2^\circ \rightarrow$ Ketones).
  - Dehydrogenation of alcohols.
  - From Alkenes: Ozonolysis.
  - **For Aldehydes:** Rosenmund reduction ($RCOCl \xrightarrow{H_2/Pd-BaSO_4} RCHO$), Stephen reaction ($R-CN \xrightarrow{SnCl_2/HCl} RCH=NH \xrightarrow{H_3O^+} RCHO$), Etard reaction (Toluene $\rightarrow$ Benzaldehyde), Gattermann-Koch reaction (Benzene $\rightarrow$ Benzaldehyde).
  - **For Ketones:** From acyl chlorides ($RCOCl \xrightarrow{R_2'Cd} RCOR'$), Friedel-Crafts acylation.
- **Physical Properties:** Polar carbonyl group, higher boiling points than hydrocarbons, lower than alcohols.
- **Chemical Reactions:**
  - **Nucleophilic Addition Reactions:** Characteristic reaction.
    - Addition of HCN (cyanohydrins), $NaHSO_3$, Grignard reagent, Alcohols (acetals/ketals), Ammonia derivatives.
  - **Reduction:**
    - To alcohols ($NaBH_4$, $LiAlH_4$).
    - To hydrocarbons: Clemmensen reduction ($Zn-Hg/HCl$), Wolff-Kishner reduction (NH_2NH_2/KOH).
  - **Oxidation:** Aldehydes easily oxidized (Tollens' test, Fehling's test), Ketones resist oxidation.
  - **Aldol Condensation:** Aldehydes/ketones with α-H atoms.
  - **Cannizzaro Reaction:** Aldehydes without α-H atoms.
  - **Haloform Reaction:** Methyl ketones ($RCOCH_3$) or compounds with $CH_3CH(OH)-$ group.

2. Carboxylic Acids

- **Definition:** Contain the carboxyl group (-COOH).
- **Preparation:**
  - Oxidation of 1° alcohols and aldehydes.
  - From Nitriles and Amides: Hydrolysis ($RCN \rightarrow RCOOH$).
  - From Grignard reagents: $RMgX + CO_2 \rightarrow RCOOMgX \rightarrow RCOOH$.
  - From Acyl halides and Anhydrides: Hydrolysis.
- **Physical Properties:** Strong intermolecular H-bonding, higher boiling points than alcohols.
- **Chemical Reactions:**
  - **Acidity:** More acidic than phenols and alcohols. e^- withdrawing groups increase acidity.
  - **Reactions involving -OH group:**
    - Formation of Anhydrides, Esters (Esterification), Acyl chlorides.
  - **Reactions involving -COOH group:**
    - Reduction to alcohols ($LiAlH_4$).
    - Decarboxylation ($RCOOH \xrightarrow{NaOH/CaO} RH$).
  - **Hell-Volhard-Zelinsky (HVZ) Reaction:** Halogenation at α-carbon.

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Amines

- **Definition:** Derivatives of ammonia where one or more H atoms are replaced by alkyl or aryl groups.
- **Classification:** Primary (1°), Secondary (2°), Tertiary (3°).
- **Preparation:**
  - Reduction of nitro compounds ($RNO_2 \xrightarrow{H_2/Pd} RNH_2$).
  - Ammonolysis of alkyl halides ($RX \xrightarrow{NH_3} RNH_2$).
  - Reduction of nitriles ($RCN \xrightarrow{LiAlH_4} RCH_2NH_2$).
  - Reduction of amides ($RCONH_2 \xrightarrow{LiAlH_4} RCH_2NH_2$).
  - Gabriel Phthalimide Synthesis (1° amines only, $ArNH_2$ not formed).
  - Hoffmann Bromamide Degradation ($RCONH_2 \xrightarrow{Br_2/NaOH} RNH_2 + Na_2CO_3 + NaBr + H_2O$).
- **Physical Properties:**
  - Lower amines are gases/liquids, higher are solids.
  - 1° and 2° amines form H-bonds, higher boiling points than non-polar compounds.
  - Soluble in water (short chain).
- **Chemical Reactions:**
  - **Basicity:** Due to lone pair on N. $2^\circ > 1^\circ > 3^\circ$ (aqueous solution, alkyl amines), $3^\circ > 2^\circ > 1^\circ$ (gaseous state). Aromatic amines are less basic than alkyl amines.
  - **Acylation:** $RNH_2 + CH_3COCl \rightarrow RNHCOCH_3 + HCl$.
  - **Carbylamine Reaction:** Test for 1° amines ($RNH_2 \xrightarrow{CHCl_3/KOH} RNC$ (foul smell)).
  - **Reaction with Nitrous Acid (HNO_2):**
    - 1° aliphatic: $RNH_2 \rightarrow$ alcohol + N_2.
    - 1° aromatic: $ArNH_2 \rightarrow$ Diazonium salt.
  - **Hinsberg Test:** To distinguish $1^\circ, 2^\circ, 3^\circ$ amines using benzene sulphonyl chloride.
  - **Electrophilic Substitution (Aromatic Amines):** $-NH_2$ is strongly activating and o,p-directing.
    - Bromination, Nitration, Sulphonation.

Biomolecules

1. Carbohydrates

- **Definition:** Polyhydroxy aldehydes or ketones. General formula $C_x(H_2O)_y$.
- **Classification:**
  - **Monosaccharides:** Glucose, Fructose.
  - **Disaccharides:** Sucrose (Glucose + Fructose), Maltose (Glucose + Glucose), Lactose (Glucose + Galactose).
  - **Polysaccharides:** Starch, Cellulose, Glycogen.
- **Glucose:** Aldohexose, ring structure (pyranose), reducing sugar.
- **Fructose:** Ketohexose, ring structure (furanose), reducing sugar.
- **Sucrose:** Non-reducing sugar.
- **Starch:** Amylose (linear, α-1,4 linkages) + Amylopectin (branched, α-1,4 and α-1,6 linkages).
- **Cellulose:** Linear polymer of β-D-glucose.

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2. Proteins

- **Definition:** Polymers of α-amino acids linked by peptide bonds.
- **Amino Acids:** Contain both amino ($-NH_2$) and carboxyl (-COOH) groups. Zwitterionic form.
- **Peptide Bond:** $-CO-NH-$.
- **Structure:**
  - **Primary:** Sequence of amino acids.
  - **Secondary:** α-helix, β-pleated sheet (H-bonding).
  - **Tertiary:** 3D folding (H-bonds, ionic, disulfide, hydrophobic interactions).
  - **Quaternary:** Arrangement of multiple polypeptide units.
- **Denaturation:** Loss of biological activity due to change in $2^\circ, 3^\circ, 4^\circ$ structures (heat, pH change).

3. Vitamins

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- **Definition:** Organic compounds required in small amounts for normal growth and metabolic functions.
- **Classification:**

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  - **Fat-soluble:** A, D, E, K.
  - **Water-soluble:** B-complex, C.

4. Nucleic Acids

- **Definition:** DNA (Deoxyribonucleic acid) and RNA (Ribonucleic acid). Genetic material.
- **Components:**
  - **Pentose sugar:** Deoxyribose (DNA), Ribose (RNA).
  - **Nitrogenous bases:**
    - Purines: Adenine (A), Guanine (G).
    - Pyrimidines: Cytosine (C), Thymine (T) in DNA; Cytosine (C), Uracil (U) in RNA.
  - **Phosphate group.**
- **Structure:**
  - **DNA:** Double helix, A-T and G-C base pairing.
  - **RNA:** Single strand, various types (mRNA, tRNA, rRNA).

5. Hormones

- **Definition:** Chemical messengers secreted by endocrine glands.
- **Types:** Steroids, peptides, amino acid derivatives.

Coordination Compounds

- **Definition:** Compounds containing a central metal atom/ion bonded to a number of ions or molecules (ligands).
- **Key Terms:**
  - **Central Metal Ion:** Lewis acid.
  - **Ligand:** Lewis base (donates lone pair). Monodentate, bidentate, polydentate.
  - **Coordination Number:** Number of ligand donor atoms directly bonded to the central metal ion.
  - **Coordination Sphere:** Central metal ion + ligands.
  - **Counter Ion:** Ion outside the coordination sphere.
- **Nomenclature:** IUPAC rules.
- **Isomerism:**
  - **Structural Isomerism:**
    - Ionization: $[Co(NH_3)_5Br]SO_4$ and $[Co(NH_3)_5SO_4]Br$.
    - Hydrate: $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$.
    - Linkage: $[Co(NH_3)_5ONO]Cl_2$ and $[Co(NH_3)_5NO_2]Cl_2$.
    - Coordination: $[Co(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$.
  - **Stereoisomerism:**
    - Geometric (cis-trans): For square planar (MA_2B_2, MA_2BC) and octahedral (MA_4B_2, MA_3B_3, $M(AA)_2B_2$).
    - Optical (enantiomers): Non-superimposable mirror images. Common in octahedral complexes (e.g., $M(AA)_3$, $M(AA)_2B_2$ (cis form)).
- **Bonding Theories:**
  - **Werner's Theory:** Primary (ionizable) and Secondary (non-ionizable) valencies.
  - **Valence Bond Theory (VBT):** Explains geometry and magnetic properties (hybridization: $sp^3, dsp^2, d^2sp^3, sp^3d^2$).
  - **Crystal Field Theory (CFT):** Explains splitting of d-orbitals in presence of ligands.
    - Octahedral splitting (Δ_o), Tetrahedral splitting (Δ_t).
    - Strong field ligands (large splitting, low spin), Weak field ligands (small splitting, high spin).
    - Colour of complexes.
- **Stability of Coordination Compounds:** Formation constant (K_f). Chelate effect.

D and F Block Elements

1. D-block Elements (Transition Elements)

- **Definition:** Elements with incompletely filled d-orbitals in their ground state or in any of their common oxidation states. (Zn, Cd, Hg are not considered transition elements).
- **Electronic Configuration:** $(n-1)d^{1-10}ns^{1-2}$.
- **General Characteristics:**
  - **Metallic Character:** High melting/boiling points, good conductors.
  - **Variable Oxidation States:** Due to participation of $(n-1)d$ and ns electrons. Max OS in middle of series.
  - **Formation of Coloured Ions:** Due to d-d transitions.
  - **Formation of Complex Compounds:** Due to small size, high charge, and vacant d-orbitals.
  - **Catalytic Properties:** Due to variable oxidation states and large surface area.
  - **Paramagnetism:** Due to unpaired electrons.
  - **Formation of Interstitial Compounds:** Small atoms (H, C, N) trapped in voids.
  - **Alloy Formation:** Similar atomic radii.
- **Important Compounds:**
  - **Potassium Dichromate ($K_2Cr_2O_7$):** Orange, strong oxidizing agent in acidic medium ($Cr_2O_7^{2-} \xrightarrow{H^+} Cr^{3+}$).
  - **Potassium Permanganate ($KMnO_4$):** Purple, strong oxidizing agent.
    - Acidic: $MnO_4^- \xrightarrow{H^+} Mn^{2+}$
    - Neutral: $MnO_4^- \xrightarrow{H_2O} MnO_2$
    - Alkaline: $MnO_4^- \xrightarrow{OH^-} MnO_4^{2-}$

2. F-block Elements (Inner Transition Elements)

- **Definition:** Elements in which the 4f (Lanthanoids) or 5f (Actinoids) orbitals are being filled.

a. Lanthanoids (4f series)

- **Electronic Configuration:** $[Xe]4f^{1-14}5d^{0-1}6s^2$.
- **Oxidation State:** Primarily +3, sometimes +2, +4.
- **Lanthanoid Contraction:** Steady decrease in atomic/ionic radii with increasing atomic number.
  - Causes similarity in properties of 2nd and 3rd transition series elements.
- **General Characteristics:**
  - Silvery-white soft metals.
  - Coloured ions (due to f-f transitions).
  - Paramagnetic.
  - Chemical reactivity similar.

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b. Actinoids (5f series)

- **Electronic Configuration:** $[Rn]5f^{1-14}6d^{0-1}7s^2$.
- **Oxidation State:** Primarily +3, but also +4, +5, +6, +7 (more variable than lanthanoids).
- **Actinoid Contraction:** Similar to lanthanoid contraction, but more pronounced.
- **General Characteristics:**
  - Radioactive.
  - Tendency to form complexes.
  - More reactive than lanthanoids.
  - Coloured ions.

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Electrochemistry

- **Definition:** Study of production of electricity from energy released during spontaneous chemical reactions and use of electrical energy to bring about non-spontaneous chemical transformations.

1. Electrochemical Cells (Galvanic/Voltaic Cells)

- **Principle:** Convert chemical energy into electrical energy (spontaneous redox reactions).
- **Components:** Two half-cells (anode, cathode), salt bridge.
- **Anode:** Oxidation occurs (negative electrode).
- **Cathode:** Reduction occurs (positive electrode).
- **Salt Bridge:** Maintains electrical neutrality.
- **Cell Notation:** Anode | Anode ion || Cathode ion | Cathode.
- **Cell Potential (EMF):** $E_{cell} = E_{cathode} - E_{anode}$.
- **Standard Hydrogen Electrode (SHE):** Reference electrode (0.0 V).

2. Nernst Equation

- Relates cell potential to concentrations of reactants and products: $E_{cell} = E^\circ_{cell} - \frac{RT}{nF}\ln Q$ At $298K$: $E_{cell} = E^\circ_{cell} - \frac{0.0591}{n}\log Q$
- **Relationship with Gibbs Energy:** $\Delta G = -nFE_{cell}$, $\Delta G^\circ = -nFE^\circ_{cell}$.
- **Relationship with Equilibrium Constant:** $\Delta G^\circ = -RT\ln K_{eq}$, $E^\circ_{cell} = \frac{RT}{nF}\ln K_{eq}$.

3. Electrolytic Cells

- **Principle:** Convert electrical energy into chemical energy (non-spontaneous reactions).
- **Anode:** Oxidation occurs (positive electrode).
- **Cathode:** Reduction occurs (negative electrode).
- **Electrolysis:** Decomposition of electrolyte by electric current.
- **Faraday's Laws of Electrolysis:**
  - **First Law:** $W \propto Q$ ($W = ZIt$).
  - **Second Law:** $W_1/W_2 = E_1/E_2$.

4. Conductance of Electrolytic Solutions

- **Conductance (G):** Reciprocal of resistance ($G = 1/R$). Unit: Siemens (S).
- **Conductivity (κ):** Conductance of a unit volume. $\kappa = G \times l/A$. Unit: S cm^{-1}.
- **Molar Conductivity (Λ_m):** $\Lambda_m = \frac{\kappa \times 1000}{C}$. Unit: S cm^2 mol^{-1}.
- **Variation of Conductivity with Concentration:**
  - Strong electrolytes: Λ_m increases slightly with dilution (Debye-Huckel-Onsager equation).
  - Weak electrolytes: Λ_m increases significantly with dilution (ionization increases).
- **Kohlrausch's Law:** At infinite dilution, $\Lambda^\circ_m = x\lambda^\circ_A + y\lambda^\circ_B$.
  - Applications: Calculation of Λ°_m for weak electrolytes, calculation of degree of dissociation ($\alpha = \Lambda_m/\Lambda^\circ_m$).

5. Batteries and Fuel Cells

- **Primary Batteries:** Non-rechargeable (e.g., Dry cell, Mercury cell).
- **Secondary Batteries:** Rechargeable (e.g., Lead-acid battery, Ni-Cd cell).
- **Fuel Cells:** Convert energy of combustion of fuels directly into electrical energy (e.g., H_2-O_2 fuel cell).
- **Corrosion:** Electrochemical phenomenon (e.g., rusting of iron).

Chemical Kinetics

- **Definition:** Study of reaction rates, factors affecting rates, and reaction mechanisms.

1. Rate of Reaction

- **Average Rate:** Change in concentration / Change in time.
- **Instantaneous Rate:** Rate at a particular instant.
- **Units:** mol L^{-1} s^{-1}.
- **Stoichiometry:** For $aA + bB \rightarrow cC + dD$, Rate $= -\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = \frac{1}{c}\frac{d[C]}{dt} = \frac{1}{d}\frac{d[D]}{dt}$.

2. Factors Affecting Rate

- Concentration of reactants.
- Temperature.

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- Catalyst.
- Surface area (for heterogeneous reactions).
- Nature of reactants.

3. Rate Law and Order of Reaction

- **Rate Law:** Rate $= k[A]^x[B]^y$.
- **Order of Reaction:** Sum of powers of concentrations in rate law ($x+y$). Can be 0, 1, 2, fractional.
- **Molecularity:** Number of reacting species in an elementary step. Always a whole number, ≤ 3.
- **Elementary Reaction:** Occurs in one step.
- **Complex Reaction:** Occurs in multiple steps. Slowest step is rate-determining.

4. Integrated Rate Equations

- **Zero Order:** $k = \frac{[R]*0 - [R]*}{t}$.$Half-life(t{1/2})=$ \frac{[R]_0}{2k}$.
- **First Order:** $k = \frac{2.303}{t} \log \frac{[R]*0*}{[R]}$.$Half-life(t{1/2})=$ \frac{0.693}{k}$.
- **Second Order:** $k = \frac{1}{t} \left(\frac{1}{[R]} - \frac{1}{[R]*0* \right)$.$Half-life(t{1/2})=$ \frac{1}{k[R]_0}$.
- **Pseudo First Order Reaction:** Appears first order but is actually higher order (e.g., hydrolysis of ester in excess water).

5. Temperature Dependence of Rate

- **Arrhenius Equation:** $k = Ae^{-E_a/RT}$.
- $\ln k = \ln A - \frac{E_a}{RT}$.
- $\log\frac{k_2}{k_1} = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1T_2}\right)$.
- **Activation Energy (E_a):** Minimum energy required for reactants to form products.
- **Collision Theory:** Reactions occur due to effective collisions.

6. Catalysis

- **Catalyst:** Substance that alters the rate of a reaction without being consumed.
- Provides an alternative reaction pathway with lower activation energy.
- Does not change ΔG or equilibrium constant.
- **Homogeneous Catalysis:** Catalyst and reactants in same phase.
- **Heterogeneous Catalysis:** Catalyst and reactants in different phases.

Solutions

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- **Definition:** Homogeneous mixture of two or more components.
- **Components:** Solvent (larger quantity), Solute (smaller quantity).

1. Types of Solutions

- Solid in solid, solid in liquid, liquid in liquid, gas in liquid, gas in gas, etc.

2. Concentration Terms

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- **Mass %:** $\frac{\text{mass of solute}}{\text{mass of solution}} \times 100$.
- **Volume %:** $\frac{\text{volume of solute}}{\text{volume of solution}} \times 100$.
- **Mass by Volume %:** $\frac{\text{mass of solute}}{\text{volume of solution}} \times 100$.
- **Parts Per Million (ppm):** $\frac{\text{mass of solute}}{\text{mass of solution}} \times 10^6$.
- **Mole Fraction (χ):** $\frac{\text{moles of component}}{\text{total moles of all components}}$.
- **Molarity (M):** $\frac{\text{moles of solute}}{\text{volume of solution (L)}}$. (Temperature dependent)
- **Molality (m):** $\frac{\text{moles of solute}}{\text{mass of solvent (kg)}}$. (Temperature independent)

3. Solubility

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- **Solid in Liquid:** Increases with temperature (endothermic dissolution), decreases (exothermic dissolution).
- **Gas in Liquid:** Decreases with temperature, increases with pressure (Henry's Law: $P = K_H x$).

4. Colligative Properties (Depend on number of solute particles, not their identity)

- **Relative Lowering of Vapour Pressure ($RLVP$):**
  - Raoult's Law (for non-volatile solute): $\frac{P^\circ - P_s}{P^\circ} = \chi_{solute}$.
  - For volatile components: $P_{total} = P_A + P_B = P_A^\circ \chi_A + P_B^\circ \chi_B$.
- **Elevation in Boiling Point (ΔT_b):**
  - $\Delta T_b = K_b m$. (K_b = molal elevation constant/ebullioscopic constant).
- **Depression in Freezing Point (ΔT_f):**
  - $\Delta T_f = K_f m$. (K_f = molal depression constant/cryoscopic constant).
- **Osmotic Pressure (Π):**
  - $\Pi = CRT = \frac{n}{V} RT$.
  - Isotonic solutions have same osmotic pressure.

5. Abnormal Molar Masses and Van't Hoff Factor (i)

- **Association:** $i < 1$.
- **Dissociation:** $i > 1$.
- **Modified Colligative Properties:**
  - $RLVP = i\chi_{solute}$
  - $\Delta T_b = iK_b m$
  - $\Delta T_f = iK_f m$
  - $\Pi = iCRT$
- **Van't Hoff Factor:** $i = \frac{\text{Normal molar mass}}{\text{Abnormal molar mass}} = \frac{\text{Observed colligative property}}{\text{Calculated colligative property}}$.
- $i = 1 + (n-1)\alpha$ (for dissociation, where n is number of ions).
- $i = 1 - \frac{1}{n}(1-\alpha)$ (for association, where n is number of molecules associating).

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Formula Cheatsheet

Ideal Solution

- **Interactions:** A-B interactions are equal to A-A and B-B interactions.
- **Properties:**
  - $\Delta H_{mix} = 0$
  - $\Delta V_{mix} = 0$
- **Raoult's Law:** $P_A = P_A^0 X_A$ and $P_B = P_B^0 X_B$
- **Total Pressure:** $P_{total} = P_A^0 X_A + P_B^0 X_B$
- **Example:** n-Hexane + n-Heptane

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Non-Ideal Solution

Positive Deviation

- **Interactions:** A-B interactions are weaker than A-A and B-B interactions.
- **Properties:**
  - $\Delta H_{mix} = +ve$
  - $\Delta V_{mix} = +ve$
  - $P_{total} > P_A^0 X_A + P_B^0 X_B$
- **Azeotropes:** Forms minimum boiling azeotropes.
- **Example:** Acetone + Ethanol

Negative Deviation

- **Interactions:** A-B interactions are stronger than A-A and B-B interactions.
- **Properties:**
  - $\Delta H_{mix} = -ve$
  - $\Delta V_{mix} = -ve$
  - $P_{total} < P_A^0 X_A + P_B^0 X_B$
- **Azeotropes:** Forms maximum boiling azeotropes.
- **Example:** Acetone + Chloroform

Concentration Terms

- **Mass Percentage (w/w):**
  $$\frac{w_2}{w_1 + w_2} \times 100$$
- **Volume Percentage (v/v):**
  $$\frac{V_2}{V_1 + V_2} \times 100$$
- **Mass by Volume Percentage (w/v):**
  $$\frac{w_2(\text{g})}{V_{\text{solution}}(\text{mL})} \times 100$$
- **Mass Fraction:**
  $$X_1 = \frac{w_1}{w_1 + w_2}, \quad X_2 = \frac{w_2}{w_1 + w_2}$$
- **Parts per Million (ppm):**
  $$\frac{\text{Mass of solute}}{\text{Mass of solution}} \times 10^6$$
- **Molarity (M):**
  $$\frac{\text{Moles of solute}}{\text{Volume of solution (L)}} \quad \text{Unit: mol/L}$$
  - Inversely proportional to temperature.
- **Molality (m):**
  $$\frac{\text{Moles of solute}}{\text{Mass of solvent (kg)}} \quad \text{Unit: mol/kg}$$
  - Independent of temperature.
- **Mole Fraction (X):**
  $$X_A = \frac{n_A}{n_A + n_B}, \quad X_B = \frac{n_B}{n_A + n_B}$$
  - $X_A + X_B = 1$
  - Unitless quantity.

Colligative Properties

- Depend on the number of solute particles.

Relative Lowering of Vapor Pressure (RLVP)

- $$\frac{P^0 - P_s}{P^0} = X_{\text{solute}} = \frac{n_{\text{solute}}}{n_{\text{solute}} + n_{\text{solvent}}}$$
- For dilute solutions:
  $$\frac{P^0 - P_s}{P^0} \approx \frac{n_{\text{solute}}}{n_{\text{solvent}}}$$

Elevation in Boiling Point (ΔT_b)

- $$\Delta T_b = T_b - T_b^0 = K_b \cdot m$$
  - T_b: Boiling point of solution
  - T_b^0: Boiling point of pure solvent
  - K_b: Molal elevation constant (Ebullioscopic constant), Unit: K kg mol⁻¹

Depression in Freezing Point (ΔT_f)

- $$\Delta T_f = T_f^0 - T_f = K_f \cdot m$$
  - T_f: Freezing point of solution
  - T_f^0: Freezing point of pure solvent
  - K_f: Molal depression constant (Cryoscopic constant), Unit: K kg mol⁻¹

Osmotic Pressure (Π)

- $$\Pi = CRT = \frac{n}{V}RT = \frac{w_2}{M_2 V}RT$$
  - Π: Osmotic pressure (atm/bar)
  - C: Molar concentration (mol/L)
  - R: Gas constant (0.0821 L atm mol⁻¹ K⁻¹ or 8.314 J mol⁻¹ K⁻¹)
  - T: Temperature (K)

Van't Hoff Factor (i)

- $$i = \frac{\text{Normal Molecular Mass}}{\text{Observed Molecular Mass}} = \frac{\text{Observed Colligative}}{\text{Calculated Colligativ}}$$
- **Dissociation:** $i > 1$ (e.g., NaCl, $i = 2$)
  - Degree of dissociation (α):
  $$\alpha = \frac{i-1}{n-1}$$
- **Association:** $i < 1$ (e.g., Ethanoic acid in benzene, $i = 0.5$)
  - Degree of association (α):
  $$\alpha = \frac{1-i}{1-1/n}$$
- **No dissociation/association:** $i = 1$ (e.g., Glucose, Urea, Sucrose)
- **Modified Colligative Properties:**
  - RLVP:
  $$\frac{P^0 - P_s}{P^0} = iX_{\text{solute}}$$
  - $\Delta T_b = iK_b m$
  - $\Delta T_f = iK_f m$
  - $\Pi = iCRT$

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Henry's Law

- **Partial pressure of gas:** $p = K_H X$
  - K_H: Henry's Law constant
  - X: Mole fraction of the gas in the solution
- **Higher K_H means lower solubility.**

Electrochemistry Basics

- **Cell:** Converts one form of energy into another.
  - **Electrochemical (Galvanic/Voltaic) Cell:** Chemical energy $\rightarrow$ Electrical energy.
  - **Electrolytic Cell:** Electrical energy $\rightarrow$ Chemical energy.
- **Representation of a Cell (e.g., Daniell Cell):**
  $$Zn(s)|Zn^{2+}(aq, 1M)||Cu^{2+}(aq, 1M)|Cu(s)$$
  - Single vertical line: Phase boundary
  - Double vertical line: Salt bridge
  - Anode (oxidation) on left, Cathode (reduction) on right.

Nernst Equation

- **For a general reaction:** $aA + bB \rightleftharpoons cC + dD$
  - $$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{RT}{nF}\ln Q$$
  - $$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.0591}{n}\log Q \quad \text{(at 298K)}$$
  - $Q = \frac{[C]^c[D]^d}{[A]^a[B]^b}$ (Reaction Quotient)
  - n: Number of electrons transferred
  - F: Faraday constant (96485 C/mol)
- **At Equilibrium:** $E_{\text{cell}} = 0$
  - $$E_{\text{cell}}^0 = \frac{0.0591}{n}\log K_c \quad \text{(at 298K)}$$
  - K_c: Equilibrium constant

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Gibbs Free Energy

- $$\Delta G = -nFE_{\text{cell}}$$
- $$\Delta G^0 = -nFE_{\text{cell}}^0$$
- $$\Delta G^0 = -2.303RT\log K_c$$

Faraday's Laws of Electrolysis

First Law

- The mass of a substance deposited or liberated at any electrode is directly proportional to the quantity of electricity passed.
  - $W = Z \times I \times t = \frac{E}{F} \times I \times t$
  - Z: Electrochemical equivalent
  - E: Equivalent mass
  - I: Current (Amperes)
  - t: Time (seconds)

Second Law

- When the same quantity of electricity is passed through different electrolytes connected in series, the masses of the substances produced at the electrodes are directly proportional to their equivalent masses.
  - $$\frac{W_1}{W_2} = \frac{E_1}{E_2}$$

Conductivity of Ionic Solutions

- **Conductance (G):** Reciprocal of resistance.
  - $G = \frac{1}{R} = k\frac{A}{l}$
  - Unit: Siemens (S) or ohm⁻¹
  - Increases on dilution due to larger number of ions.
- **Specific Conductance (Conductivity, κ):**
  - $\kappa = \frac{1}{R} \times \frac{l}{A} = G \times G^*$
  - $G^* = \frac{l}{A}$ (Cell constant)
  - Unit: S cm⁻¹ or S m⁻¹
  - Decreases on dilution as the number of ions per cm³ decreases.
- **Molar Conductivity (Λ_m):**
  - $$\Lambda_m = \frac{\kappa \times 1000}{M} \quad (\text{if } \kappa \text{ in S cm}^{-1})$$
  - Unit: S cm² mol⁻¹
  - Increases with dilution due to a large increase in V (volume containing one mole of electrolyte).
- **Limiting Molar Conductivity (Λ_m^0 or Λ_0):**
  - Molar conductivity when the concentration of electrolyte approaches zero (infinite dilution).

Kohlrausch's Law

- **For an electrolyte at infinite dilution:**
  - $$\Lambda_m^0 = \nu_+\lambda_+^0 + \nu_-\lambda_-^0$$
  - ν_+ and ν_-: Number of cations and anions respectively
  - λ_+^0 and λ_-^0: Limiting molar conductivities of the cation and anion.
- **Degree of Dissociation (α):**
  - $$\alpha = \frac{\Lambda_m}{\Lambda_m^0}$$
- **Dissociation Constant (K_c):**
  - For a weak electrolyte $AB \rightleftharpoons A^+ + B^-$
  - $$K_c = \frac{C\alpha^2}{1-\alpha}$$

Debye-Hückel-Onsager Equation

- **For strong electrolytes:**
  - $$\Lambda_m = \Lambda_m^0 - A\sqrt{C}$$
  - A: Constant depending on solvent and temperature.
  - Plot of Λ_m vs $\sqrt{C}$ is a straight line with intercept Λ_m^0 and slope -A.

Batteries

Dry Cell (Leclanché Cell)

- **Anode:** Zinc container ($Zn \rightarrow Zn^{2+} + 2e^-$)
- **Cathode:** Carbon rod surrounded by MnO_2 and carbon powder.
  - $(2MnO_2 + 2NH_4^+ + 2e^- \rightarrow Mn_2O_3 + 2NH_3 + H_2O)$
- **Overall Reaction:** $Zn + 2NH_4^+ + 2MnO_2 \rightarrow Zn^{2+} + Mn_2O_3 + 2NH_3 + H_2O$

Lead Storage Battery

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- **Anode:** Lead ($Pb + SO_4^{2-} \rightarrow PbSO_4 + 2e^-$)
- **Cathode:** Lead dioxide grid ($PbO_2 + SO_4^{2-} + 4H^+ + 2e^- \rightarrow PbSO_4 + 2H_2O$)
- **Overall Reaction:** $Pb + PbO_2 + 2H_2SO_4 \rightleftharpoons 2PbSO_4 + 2H_2O$

Fuel Cell (e.g., H_2-O_2 Fuel Cell)

- **Anode:** $2H_2 + 4OH^- \rightarrow 4H_2O + 4e^-$
- **Cathode:** $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$
- **Overall Reaction:** $2H_2 + O_2 \rightarrow 2H_2O$

Corrosion of Iron (Rusting)

- **At Anode (Iron):** $2Fe \rightarrow 2Fe^{2+} + 4e^-$
- **At Cathode (Oxygen):** $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$
- **Overall Reaction:** $2Fe + O_2 + 4H^+ \rightarrow 2Fe^{2+} + 2H_2O$
- **Rust Formation:** $2Fe^{2+} + \frac{1}{2}O_2 + (2+x)H_2O \rightarrow Fe_2O_3 \cdot xH_2O + 4H^+$
  - Formula of rust: $Fe_2O_3 \cdot xH_2O$ (Hydrated Ferric Oxide)

Rate of Reaction

- **For a reaction:** $aA + bB \rightarrow cC + dD$
  - **Rate:**

$$\text{Rate} = -\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = +\frac{1}{c}\frac{d[C]}{dt} = +\frac{1}{d}\frac{d[D]}{dt}$$

- **Units of Rate:** mol L^{-1} s^{-1} or mol L^{-1} min^{-1}
- **Average Rate:** Change in concentration over a large time interval.
- **Instantaneous Rate:** Change in concentration at a specific instant of time.

Rate Law

- **For a reaction:** $aA + bB \rightarrow \text{Products}$
  - **Rate:**

$$\text{Rate} = k[A]^x[B]^y$$

  - k: Rate constant (specific rate constant)
  - x, y: Order of reaction with respect to A and B (experimentally determined)
  - **Overall Order:** $n = x + y$

Order of Reaction

Zero Order Reaction

- **Rate Law:** Rate $= k[A]^0 = k$
- **Integrated Rate Law:** $[A]_t = -kt + [A]_0$
- **Half-life ($t_{1/2}$):**

$$t_{1/2} = \frac{[A]_0}{2k}$$

- **Units of k:** mol L^{-1} s^{-1}

First Order Reaction

- **Rate Law:** Rate $= k[A]^1$
- **Integrated Rate Law:** $\ln[A]_t = -kt + \ln[A]_0$ or $k = \frac{2.303}{t}\log\frac{[A]_0}{[A]_t}$
- **Half-life ($t_{1/2}$):**

$$t_{1/2} = \frac{0.693}{k}$$

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- **Units of k:** s^{-1}

Pseudo First Order Reaction

- Reactions that are not truly first order but become first order under certain conditions (e.g., hydrolysis of ester in excess water).

Collision Theory

- **Collision Frequency (Z):** Number of collisions per second per unit volume.
- **Rate:**

$$\text{Rate} = PZ_{AB}e^{-E_a/RT}$$

  - P: Probability factor (steric factor)
  - Z_{AB}: Collision frequency for reactants A and B
  - E_a: Activation energy
  - R: Gas constant
  - T: Temperature

Arrhenius Equation

-

$$k = Ae^{-E_a/RT}$$

- **Logarithmic Form:**

$$\ln k = \ln A - \frac{E_a}{RT}$$

or

$$\log k = \log A - \frac{E_a}{2.303RT}$$

- **For two different temperatures:**

$$\log\frac{k_2}{k_1} = \frac{E_a}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

- k: Rate constant
- A: Arrhenius factor (Pre-exponential factor/Frequency factor)
- E_a: Activation energy
- R: Gas constant
- T: Temperature (K)

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Role of Catalyst

- A chemical substance that alters the rate of reaction without undergoing any chemical change.
- **Mechanism:** Provides an alternative reaction path with lower activation energy (E_a).
  - Graph: Energy vs Reaction Coordinate. Catalyst lowers the peak (activation energy) of the reaction pathway.

d- and f-Block Elements

General Electronic Configuration

- **d-block:** $(n-1)d^{1-10}ns^{0-2}$
- **f-block:** $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$

Properties of Transition Elements (d-block)

- **Melting & Boiling Points:** High due to strong metallic bonding (unpaired d-electrons).
- **Enthalpies of Atomization:** High due to strong interatomic interactions.
- **Oxidation States:** Variable oxidation states due to participation of ns and (n-1)d electrons. (Highest in 3d series: Mn, +7).
- **Atomic Radii:** Decreases from left to right, then remains almost constant in the middle, and increases towards the end of the series.
- **Complex Formation:** Form complexes due to high nuclear charge, small size, and availability of empty d-orbitals to accept lone pairs from ligands.
- **Coloured Compounds:** Form coloured compounds due to d-d transitions (presence of unpaired d-electrons).
- **Alloy Formation:** Due to similar atomic radii, atoms of one metal can easily replace atoms of another.
- **Interstitial Compounds:** Small atoms (H, C, N) can be accommodated in the interstitial spaces of their lattices.
- **Magnetic Properties:**
  - **Paramagnetic:** Due to unpaired d-electrons. Calculated using $\mu = \sqrt{n(n+2)}$ BM, where n is the number of unpaired electrons.
  - **Diamagnetic:** All electrons are paired.
- **Oxides:**
  - Higher oxidation state oxides are generally acidic (e.g., CrO_3).
  - Lower oxidation state oxides are basic (e.g., CrO).
  - Intermediate oxidation state oxides are amphoteric (e.g., Cr_2O_3).

Potassium Dichromate ($K_2Cr_2O_7$)

- **Structure:**
  - Dichromate ion ($Cr_2O_7^{2-}$): Orange, tetrahedral units sharing one oxygen.
  - Chromate ion (CrO_4^{2-}): Yellow, tetrahedral.
  - Equilibrium: $Cr_2O_7^{2-} + H_2O \rightleftharpoons 2CrO_4^{2-} + 2H^+$ (pH dependent)
- **Preparation:** From chromite ore ($FeCr_2O_4$).
  - $4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \xrightarrow{\text{Fusion}} 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2$
  - $2Na_2CrO_4 + 2H^+ \rightarrow Na_2Cr_2O_7 + 2Na^+ + H_2O$
  - $Na_2Cr_2O_7 + 2KCl \rightarrow K_2Cr_2O_7 + 2NaCl$
- **Oxidizing Action (Acidic Medium):**
  - $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$
  - Oxidizes Fe^{2+} to Fe^{3+}, I^- to I_2, S^{2-} to S, SO_3^{2-} to SO_4^{2-}, etc.

Potassium Permanganate ($KMnO_4$)

- Deep purple crystalline solid, strong oxidizing agent.
- **Preparation:** From pyrolusite ore (MnO_2).
  - $2MnO_2 + 4KOH + O_2 \xrightarrow{\text{Fusion}} 2K_2MnO_4 + 2H_2O$ (Green manganate)
  - $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$ (Disproportionation in acidic medium)
- **Oxidizing Action (Acidic Medium):**
  - $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$
  - Oxidizes Fe^{2+} to Fe^{3+}, $C_2O_4^{2-}$ (oxalate) to CO_2, I^- to I_2, H_2S to S, SO_2 to SO_4^{2-}, etc.
- **Oxidizing Action (Neutral/Weakly Alkaline Medium):**
  - $MnO_4^- + 2H_2O + 3e^- \rightarrow MnO_2 + 4OH^-$
- **Oxidizing Action (Strongly Alkaline Medium):**
  - $MnO_4^- + e^- \rightarrow MnO_4^{2-}$

f-Block Elements (Inner Transition Elements)

- **Lanthanoids (4f series):** Last electron enters 4f orbital.
- **Actinoids (5f series):** Last electron enters 5f orbital.

Lanthanoid Contraction

- **Definition:** Gradual and steady decrease in atomic and ionic radii from Lanthanum (La) to Lutetium (Lu) across the lanthanoid series.
- **Cause:** Poor shielding effect of 4f electrons. As atomic number increases, nuclear charge increases, but 4f electrons do not effectively shield the outer electrons, leading to increased effective nuclear charge and contraction of size.
- **Consequences:**
  - Difficulty in chemical separation of lanthanoids.
  - Similar atomic radii for elements of 2nd and 3rd transition series (e.g., Zr/Hf, Nb/Ta).
  - Basic character of hydroxides decreases from $La(OH)_3$ (most basic) to $Lu(OH)_3$ (least basic).

Differences between Lanthanoids and Actinoids

Property	Lanthanoids	Actinoids
Electronic Conf.	$4f^{1-14}5d^{0-1}6s^2$	$5f^{1-14}6d^{0-1}7s^2$
Common Oxi. State	+3 (most stable)	+3 (most common), but also +4, +5, +6, +7
Contraction	Lanthanoid contraction	Actinoid contraction (greater than lanthanoid)
Chemical Reactivity	Less reactive	More reactive
Magnetic Properties	Paramagnetic (due to unpaired 4f electrons)	Highly paramagnetic (due to unpaired 5f electrons)
Colour	Many ions are colourless	Most ions are coloured
Complex Formation	Lesser tendency to form complexes	Greater tendency to form complexes
Oxocations	Do not form oxocations	Form oxocations (e.g., UO_2^{2+}, PuO_2^{2+})
Basicity of Hydroxides	Less basic	More basic
Radioactivity	Non-radioactive (except Promethium, Pm)	All are radioactive

Coordination Compounds

Addition Compounds

- Formed by the combination of two or more simple compounds.
- **Double Salts:** Dissociate completely into constituent ions in solution (e.g., Carnallite ($KCl \cdot MgCl_2 \cdot 6H_2O$), Mohr's Salt ($FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$)).
- **Coordination Compounds:** Retain their identity in solid state and in solution (e.g., $[Co(NH_3)_6]Cl_3$).

Werner's Theory

- Metal shows two kinds of valencies:
  - **Primary Valency:**
    - Ionizable, non-directional.
    - Equal to the oxidation state of the central metal ion.
    - Satisfied by anions.
  - **Secondary Valency:**
    - Non-ionizable, directional.
    - Equal to the coordination number of the central metal ion.
    - Satisfied by ligands (anions, cations, or neutral molecules). Determines the geometry.

Ligands

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- Atoms, ions, or molecules that donate lone pairs of electrons to the central metal atom.

Classification by Charge

- **Negative Ligands:** Cl^-, F^-, CN^-, NO_2^-, OH^-, $C_2O_4^{2-}$ (oxalate), $S_2O_3^{2-}$ (thiosulphate), SCN^- (thiocyanato)
- **Neutral Ligands:** H_2O (aqua), NH_3 (ammine), CO (carbonyl), NO (nitrosyl), en (ethylenediamine)
- **Positive Ligands:** NO_2^+ (nitronium), $N_2H_5^+$ (hydrazinium) - Rare

Classification by Number of Donor Sites

- **Monodentate:** One donor site (e.g., Cl^-, NH_3, H_2O)
- **Bidentate:** Two donor sites (e.g., en, ox^{2-} (oxalate))
- **Polydentate:** More than two donor sites (e.g., EDTA^{4-} (hexadentate))

Chelating Ligands

- Bidentate or polydentate ligands that form a ring structure with the central metal ion. Chelate complexes are more stable (chelate effect).

Ambidentate Ligands

- Monodentate ligands that can coordinate through two different atoms (e.g., NO_2^- (nitro/-nitrito), SCN^- (thiocyanato/-isothiocyanato), CN^- (cyano/-isocyano)).

Isomerism in Coordination Compounds

Structural Isomerism

- Same molecular formula, different connectivity.
  - **Ionisation Isomerism:** Ligand and counter-ion exchange positions (e.g., $[Co(NH_3)_5Br]SO_4$ and $[Co(NH_3)_5SO_4]Br$).
  - **Hydrate Isomerism:** Water molecules inside and outside the coordination sphere (e.g., $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$).
  - **Linkage Isomerism:** Ambidentate ligands coordinate through different atoms (e.g., $[Co(NH_3)_5NO_2]^{2+}$ (nitro) and $[Co(NH_3)_5ONO]^{2+}$ (nitrito)).
  - **Coordination Isomerism:** Exchange of ligands between cationic and anionic coordination entities (e.g., $[Co(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$).

Stereoisomerism

- Same molecular formula and connectivity, different spatial arrangement.
  - **Geometrical Isomerism (cis-trans):**
    - **cis:** Same ligands occupy adjacent positions.
    - **trans:** Same ligands occupy opposite positions.
    - Common in square planar (MA_2X_2) and octahedral (MA_2X_4, $MA_2B_2C_2$, $M(AA)_2X_2$) complexes.
  - **Optical Isomerism (Enantiomerism):**
    - Non-superimposable mirror images (chiral molecules).
    - Optically active.
    - No plane of symmetry or center of symmetry.
    - Common in octahedral complexes ($M(AA)_3$, $M(AA)_2X_2$, $MA_2B_2C_2$ (cis form only)).
    - Example: $[Co(en)_3]^{3+}$

Valence Bond Theory (VBT)

- Metal-ligand bond is formed by overlap of empty metal orbitals with filled ligand orbitals (coordinate bond).
- Metal atom/ion undergoes hybridization to accommodate ligand electron pairs.
- **Hybridization and Geometry:**

Hybridization	Coordination Number	Geometry	Example
sp	2	Linear	$[Ag(CN)_2]^-$
sp^2	3	Trigonal planar	$[HgI_3]^-$ (rare)
sp^3	4	Tetrahedral	$[NiCl_4]^{2-}$
dsp^2	4	Square planar	$[Ni(CN)_4]^{2-}$
dsp^3 or sp^3d	5	Trigonal bipyramidal	$[Fe(CO)_5]$ (rare)
d^2sp^3	6	Octahedral (inner)	$[Co(NH_3)_6]^{3+}$
sp^3d^2	6	Octahedral (outer)	$[CoF_6]^{3-}$

- **Inner Orbital Complex (Low Spin):** Involves inner (n-1)d orbitals for hybridization (strong field ligands).
- **Outer Orbital Complex (High Spin):** Involves outer nd orbitals for hybridization (weak field ligands).
- **Magnetic Properties:** Predicted by the number of unpaired electrons.
  - **Paramagnetic:** Unpaired electrons (e.g., $[FeF_6]^{3-}$).
  - **Diamagnetic:** All electrons paired (e.g., $[Co(NH_3)_6]^{3+}$).

Crystal Field Theory (CFT)

- Metal-ligand bond is purely ionic (electrostatic interaction).
- Ligands are treated as point charges (anions) or dipoles (neutral molecules).
- Degeneracy of d-orbitals is lifted in the presence of ligand field.

Crystal Field Splitting in Octahedral Complexes

- d-orbitals split into two sets:
  - e_g **set:** $d_{x^2-y^2}, d_{z^2}$ (higher energy)
  - t_{2g} **set:** d_{xy}, d_{yz}, d_{zx} (lower energy)
- **Crystal Field Splitting Energy (Δ_o or 10 Dq):** Energy difference between e_g and t_{2g} sets.

Crystal Field Splitting in Tetrahedral Complexes

- d-orbitals split into two sets:
  - t_2 **set:** d_{xy}, d_{yz}, d_{zx} (higher energy)
  - **e set:** $d_{x^2-y^2}, d_{z^2}$ (lower energy)
- **Crystal Field Splitting Energy (Δ_t):** $\Delta_t = \frac{4}{9}\Delta_o$. Tetrahedral complexes are always high spin.

Spectrochemical Series

- Arrangement of ligands in increasing order of their field strength (ability to cause crystal field splitting):

$I^- < Br^- < S^{2-} < SCN^- < Cl^- < NO_3^- < F^- < OH$

  - **Weak Field Ligands:** Cause small splitting, form high spin complexes ($\Delta_o < P$, where P is pairing energy).
  - **Strong Field Ligands:** Cause large splitting, form low spin complexes ($\Delta_o > P$).

Colour in Coordination Compounds

- Due to d-d transitions (absorption of light in visible region, transition of electron from t_{2g} to e_g in octahedral complexes). The complementary colour is observed.

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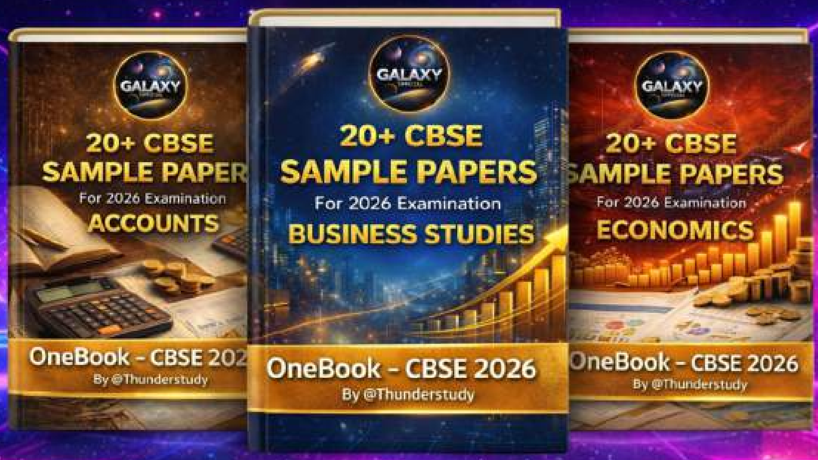

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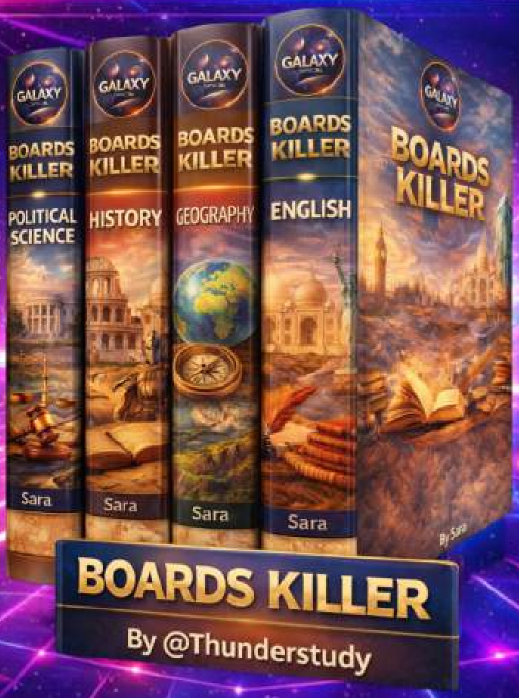

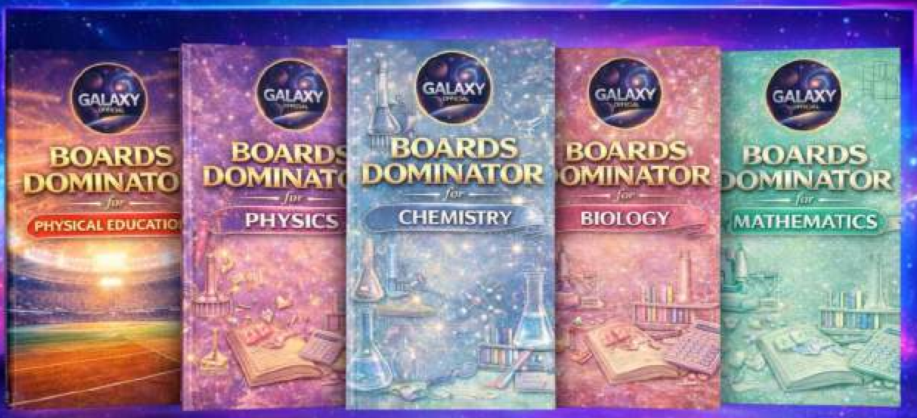

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